

Configuration Inputs

Analyst → mamplan.json

tool, files[], project_id
owner, analyst, datatype
cluster, lifetime, auth

Admin → mamplate.json

image, pullPolicy, port
resources (cpu / memory)
command with __tokens__

DevOps → config.json

clusters: {host, namespace,
kubeconfig, ingressClass}
s3: {endpoint, keys, prefix}

Mampok

CLI Interface

deploy / stop / redeploy
create-mamplan / check-status
list-expiring / stop-expired

Orchestrator

- 1. Merge Mamplan + Mamplate
- 2. Expand __token__ variables
- 3. Build Kubernetes manifests
- 4. Upload files to S3
- 5. Apply & await readiness
- 6. Write back status + URL

Infrastructure Outputs

K8s Kubernetes Cluster

Deployment · Service
Ingress · Secret
→ HTTPS URL generated

S3 Object Storage

analysis_data/ (input files)
container_data/ runtime outputs)
→ persistent across restarts

<https://cluster.institute.org/cellxgene/my-project>